

A transcendence question for a catalytic q -difference series

arising from the geodesic growth of a planar reflection group (OEIS A396406)

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Summary. A natural integer power series $F(q) \in \mathbb{Z}[[q]]$ of radius 1, defined by an explicit linear q -difference equation with the dilation $t \mapsto q^2t$, appears to be transcendental over $\mathbb{Q}(q)$, and the reduction modulo every prime p appears *maximally non- p -automatic*. All the standard transcendence tools we have tried provably cap out at a precise obstruction (a “natural-boundary” degree balance). **We are looking for a transcendence criterion, or a pointer to the right machinery or person, for this specific q -difference specialization.**

The object

Let W be the group generated by the three side-reflections of a right triangle in the plane with unequal rational legs. Its geodesic growth series $U(x) = \sum_{n \geq 0} u_n x^n$ is OEIS A396406 (u_n = number of group elements at word-length n). A catalytic (lamplighter) transfer for the underlying length statistic, with $q = x^2$ and a catalytic variable t , yields a bivariate generating function $F(q, t) = \sum_{s \geq 1} F_s(q) t^s$ satisfying the *validated* functional equation

$$F(q, t) = E(t) + D(t) F(q, q^2 t), \quad D(t) = \frac{2q(q-1)t}{(1-qt)(1-q^2t)}, \quad E(t) = \frac{2qt}{1-qt} (1 + F(q, q)). \quad (1)$$

Because the catalytic variable enters through the *dilation* $t \mapsto q^2t$, equation (1) is a linear q -difference (not polynomial-kernel / Bousquet-Mélou–Jehanne) equation; $F(q, q^2t)$ still depends on t . The series of interest is the diagonal specialization

$$F(q) := F(q, 1) = 2q + 2q^2 + 6q^3 + 2q^4 + 18q^5 + 6q^6 + 42q^7 + 18q^8 + 118q^9 + \dots \in \mathbb{Z}[[q]],$$

the “bulk-run” block of the metric; the true geodesic series U is a rational function of F and its companions. Telescoping (1) gives the closed form $F(q, t) = \sum_{k \geq 0} \alpha_k(t) E(q^{2k}t)$ with $\alpha_k(t) = [2q(q-1)]^k t^k q^{k(k-1)} / (qt; q)_{2k}$ (verified symbolically), and at $t = 1$,

$$F(q) = 2(1 + F(q, q)) \Phi(q), \quad \Phi(q) = \sum_{k \geq 0} \frac{2^k (q-1)^k q^{(k+1)^2}}{(q; q)_{2k+1}} \quad (2)$$

a partial-theta-type q -hypergeometric series. The exponent $q^{(k+1)^2}$ is a genuine theta/square structure; the Euler-product denominator $(q; q)_{2k+1} = \prod_{i=1}^{2k+1} (1 - q^i)$ spreads each theta block densely across all degrees, so F has *no lacunary gaps*.

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The question

Question 1. *Is $F(q) = F(q, 1)$ (equivalently $\Phi(q)$, equivalently $U(x)$) algebraic over $\mathbb{Q}(q)$, or transcendental?*

Characteristic- p form (Christol). For each prime p , $F \bmod p \in \mathbb{F}_p[[q]]$ is algebraic over $\mathbb{F}_p(q)$ iff the sequence of its coefficients is p -automatic, iff its p -kernel is finite. So Question 1 $\bmod p$ is the decidable-in-principle question: *is the p -kernel of $F \bmod p$ finite?*

What we know (evidence for transcendental / non-automatic)

- **Maximal non-automaticity.** For $p = 3$ the p -kernel sizes through level $k = 3$ are 1, 4, 13, 39, against the full ternary tree 1, 4, 13, 40 (deficit 1); for $p = 5$, 1, 6, 30 against 1, 6, 31. Robust across overlap thresholds; the data ceiling is $k = 3$ ($p = 3$) / $k = 2$ ($p = 5$), since $k = 4$ needs ~ 400 terms. No eventual periodicity.
- **Not rational.** Hankel determinants of $(u_n \bmod 3)$ are full-rank through size 60; Berlekamp–Massey linear complexity $\approx N/2$.
- **No low-complexity char- p Mahler relation.** Over $\mathbb{F}_3$, F satisfies no relation $\sum_{i=0}^n A_i(q) F(q^{3^i}) = 0$ of Frobenius-depth ≤ 4 and $\deg A_i \leq 32$ (nullspace search).
- **The constituent blocks are unconditionally transcendental.** The four building-block series $(\Sigma_1, \Sigma_0, S_1, S_0)$ are integer series of radius 1 with super-polynomial diagonal growth, hence have $|q| = 1$ as a natural boundary by Pólya–Carlson, hence are transcendental over $\mathbb{Q}(q)$. The difficulty is that U is a *ratio* of such blocks, in which the boundary singularities can cancel.

What provably does *not* work (and why)

Every method below caps out against the same obstruction: the support of F is *dense*, and there is an exact degree balance $\#\text{zeros} \sim \#\text{poles} \sim 2N^2$ for the order- N truncation remainder, which is the algebraic signature of a natural boundary.

- **Degree / Padé / Liouville / valuation methods** all reduce to comparing $2N^2$ against $d \cdot P(N)$ with $P(N) = 2N^2 + 2N + 1$ *exactly* (computed over $\mathbb{Q}$, $\mathbb{F}_3$, $\mathbb{F}_5$ identically): the leading $2N^2$ terms match, ruling out degree $d = 1$ (so F is non-rational) but never reaching $d \geq 2$.
- **Lacunarity / gap theorems** (Mahler-method transcendence, Adamczewski–Bell–Faverjon) require *sparse* support; the $(q; q)_{2k+1}$ denominator makes the support dense.
- **Modular forms mod p :** F is not a finite eta-quotient (log-derivative recovers a growing set of 60+ nonzero exponents).
- **Parameterized (Frobenius) difference Galois theory:** in characteristic p , $\varphi(f) = f^p$ *exactly*, so the Frobenius orbit $\{f, f^p, \dots\}$ is purely algebraic over f and φ -transcendence collapses to ordinary transcendence. The genuine σ -commuting operator is the Euler derivation $\delta = t\partial_t$, but δ yields hyper-transcendence of $F(q, t)$ in t , not algebraicity of the *value* $F(q, 1)$ over $\mathbb{F}_p(q)$ – the wrong target.

The precise ask

Is there a transcendence criterion that applies to the *specialization at a regular point* ($t = 1$, where $D(1)$ is finite and nonzero) of a σ -transcendental solution of a linear q -difference equation, when the σ -orbit of the specialization point accumulates at the singular fixed point $t = 0$? Equivalently: is the partial-theta series $\Phi(q)$ of (2) known, or amenable to a known method (Rogers–Ramanujan / partial-theta transcendence, the work of Adamczewski–Dreyfus–Hardouin and Dreyfus–Hardouin–Roques on functional relations for q -difference equations, or a $q \rightarrow 1$ confluence argument that bridges the $q = 0$ -adic and $q = 1$ -adic behaviours)? Any pointer to the right result, the right technique, or the right person would be very welcome.

A second, independent question (analytic route)

The same transcendence conclusion follows by an entirely analytic route from one asymptotic estimate for the **Hahn–Exton (third Jackson) q -Bessel function** $J_{3/2}^{(3)}(x; q)$ as $q = e^{-\tau} \rightarrow 1^-$. We have reduced that estimate, by exact identities, to a question about the q -trigonometric pair of Koornwinder–Swarttouw. Sampled along $x_n = x_0 q^{n/2}$, $J_\nu^{(3)}$ satisfies $u_{n+2} - B_n u_{n+1} + u_n = 0$ with $B_n = (q^{\nu/2} + q^{-\nu/2}) - q^{1-\nu/2} x_n^2$. Three exact, verified facts:

- *Half-integer ladder*: with $f_\nu = {}_1\varphi_1(0; q^{\nu+1}; q, qx^2)$, $f_{3/2}(xq^{1/2}) = \frac{(1-q^{1/2})(1-q^{3/2})}{q^{3/2}x^2} [f_{1/2}(x) - f_{-1/2}(x)]$ — the exact q -analogue of $j_1(X) = (\sin X/X - \cos X)/X$. So the question concerns only the q -sine/ q -cosine pair $\nu = \pm \frac{1}{2}$.
- *Exact pair Casoratian*: on the $\nu = \frac{1}{2}$ lattice, $u = J_{1/2}^{(3)}(x)$ and $v = J_{-1/2}^{(3)}(xq^{-1/4})$ solve the same recurrence with constant Casoratian $-\frac{(q^{1/2}; q)_\infty (q^{3/2}; q)_\infty}{(q; q)_\infty^2} q^{1/8} (q^{-1/4} - q^{1/4})$; the amplitude *product* of the pair is therefore pinned exactly, and only the phase offset Δ (classically $\pi/2$) remains.
- *Soft target*: the application needs only $\Delta = \frac{\pi}{2} + O(\tau^\delta)$ (equivalently, oscillatory-amplitude normalisation $1 + O(\tau^\delta)$) for *some* $\delta > 0$.

In the variable $s = \log x$, the lattice step is $\tau/2$ while the classical Bessel turning-point region has s -width $O(1)$: the recurrence is a *fine* second-order discretisation of the Bessel equation there, and the local model is classical Bessel (not q -Airy). The remaining step looks like textbook asymptotic integration — a discrete Levinson (Benzaid–Lutz) transport of the recessive solution from $x = O(\tau)$ through the turning point, with per-step residuals $O(\tau^2)$. **Question 2: has this uniform $q \rightarrow 1^-$ (equivalently, coefficient-varying-with-stepsizes) Levinson analysis for the Hahn–Exton recurrence, or the amplitude normalisation of the Koornwinder–Swarttouw q -trigonometric functions, been carried out anywhere in the literature?** Zeros and orthogonality of these functions are well studied (Koelink–Swarttouw, Abreu, Bustoz, Cardoso); we have not found the uniform confluent amplitude statement.

Either question closed makes the growth series U unconditionally transcendental. Full code, data, and the verified reductions are in the repository linked above.

1 Sharpened form of the analytic question (July 2026)

All series in the problem are now identified with Koornwinder–Swarttouw q -trigonometric functions, $\cos(z; q^2) = {}_1\phi_1(0; q; q^2, q^2 z^2)$: with $z_0 = \sqrt{2(1-q)}$, $Z = z_0/\sqrt{q}$, $q = e^{-\tau}$,

$$1 - \Sigma_1 = \cos(Z; q^2), \quad S_e = \cos(z_0; q^2), \quad P_{12} = \frac{z_0}{2} \sin(z_0; q^2) - \cos(z_0; q^2),$$

each an elementary identity (proved; verified to 60 digits). The travel poles are the solutions of $\cos(Z; q^2) = 0$. The denominator bound $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$ at every pole is proved by elementary means (annulus representation, exact Gaussian moments, discrete Gronwall; no cited asymptotic theorem). Exactly one analytic statement remains open, and it now has a one-sentence form.

Open question. Let $A(\theta) = 1/(re^{i\theta}; q)_\infty$ with $r = \sqrt{2(1-q)}e^{-\tau/2}$. Prove, with an explicit remainder uniform as $\tau \rightarrow 0^+$,

$$\mathbb{E}[\operatorname{Re} A(\theta)] = \cos(z_0; q^2) \quad \text{for } \theta \sim N(\pi/2, \tau/2) \quad (\text{this identity is proved}),$$

evaluated to relative precision $O(\tau)$: the amplitude A varies on the scale $\sqrt{\tau/2}$, which is the Gaussian's own standard deviation, so no fixed-order Laplace expansion converges relatively (measured: constant relative deficit ≈ 0.39 at every order tested). Equivalently: prove the uniform confluent expansion of $\cos(z; q^2)$ at $z = \sqrt{2(1-q)}$ to relative $O(\tau)$. This single matched-scale Gaussian convolution of an explicit q -product is the entire remaining gap between the conditional and unconditional transcendence of the series U ; the corresponding relative correction is numerically $\frac{3}{4}\tau + O(\tau^2)$ at every computed pole, so the sharp constant appears to be $\frac{3}{4}$.